

**Time:** 5 minutes**Outcome:** Distinguish nominal and real variables and analyze long-run neutrality.**Difficulty:**

Intermediate

Monetary Neutrality



THE CORE IDEA

Monetary neutrality means that a lasting money-supply change affects nominal variables but not real output or employment in the long run. Nominal values are measured in money; real values measure goods, services, or purchasing power. Proportional increases in nominal wages and prices leave the real wage unchanged.



HOW TO RECOGNIZE IT

- Separate nominal variables, measured in money, from real variables, measured in goods, services, or purchasing power.
- A lasting money-supply change can alter the price level and other nominal values in the long run.
- Under monetary neutrality, long-run real output and employment remain determined by real resources and technology.
- Do not apply long-run neutrality to the short run, when sticky prices can make money affect real activity.



WATCH OUT

Do not assume money changes real variables permanently in the long run. Monetary neutrality means money affects nominal variables in the long run, not real output or employment.



WORKED EXAMPLE: NOMINAL VERSUS REAL EFFECTS

Illustrative long-run neutrality comparison

	Before	After
Money stock	2,500	5,000
Price per unit	\$5	\$10
Nominal wage per hour	\$20	\$40
Real wage (units/hour)	4	4
Real output	100 units	100 units

Under the stated long-run neutrality assumptions, doubling money doubles prices and nominal wages. The real wage is unchanged: $\$20 / \$5 = 4$ units per hour before and $\$40 / \$10 = 4$ afterward. Real output remains 100 units. Nominal amounts change while real purchasing power and productive capacity do not.



CHECK YOURSELF

A student says, "If money is neutral, monetary policy can never affect real output." What is the best correction?

**READY?**

Return to the game and master this concept.